War Sentiment & Crude Oil Analysis

Project Process Report

Articles Analyzed: 314 | VADER-Brent Correlation: r = -0.4646

1. Problem Statement

This project investigates whether media sentiment about the US-Israel-Iran conflict correlates with Brent and WTI crude oil prices. The core hypothesis is that war-related news vocabulary - phrases like 'Iran sanctions', 'Strait of Hormuz', 'Gaza conflict' - shifts measurably in tone before or alongside oil price movements, because energy traders monitor geopolitical risk signals in real time.

Two NLP approaches are compared: VADER (rule-based lexicon scoring) which reacts to vocabulary intensity, and Twitter-RoBERTa (transformer-based contextual analysis) which understands narrative tone. Key finding: VADER detects stronger correlation (r = -0.46) than RoBERTa (r = -0.30) because war journalism uses consistently negative vocabulary that VADER scores directly, while RoBERTa classifies most objective reporting as neutral.

2. Dataset Description

Data collected entirely from free sources - no paid APIs required:

Source	Method	Articles	Date Range
Google News RSS	feedparser	350 raw	Nov 2025 - Apr 2026
Combined (deduped)	Title matching	314	Nov 2025 - Apr 2026
Crude oil prices	yfinance	123 trading days	Nov 2025 - Apr 2026

Search keywords: US Israel Iran war oil, Iran sanctions oil, Israel Iran conflict crude, strait of hormuz oil, Middle East war oil price, Iran nuclear deal oil, Gaza war oil market.

Oil tickers: Brent Crude (BZ=F) and WTI Crude (CL=F) daily closing prices.

3. Workflow Diagram

```
Google News RSS  +  NewsAPI  +  yfinance
       |                |           |
   feedparser       requests    yf.download
        \              /            |
         news_raw.csv     crude_oil_prices.csv
              |
   Text Cleaning (8 steps):
   URL strip, HTML strip, tokenize,
   filter stopwords, lemmatize
              |
         news_processed.csv
           /         \
```

```
  VADER NLP     RoBERTa NLP
(lexicon)     (transformer)
       \           /
  Daily Aggregation
        |
  Merge with Oil Prices
        |
Pearson r + Rolling Correlation
        |
13 Visualizations + PDF Report
```

4. Phase-by-Phase Explanation

Phase 1 - Data Collection (01_Data_Collection.ipynb)

Collected news articles using Google News RSS feeds via feedparser. Seven geopolitical keywords targeting the US-Israel-Iran conflict and energy markets were queried (50 articles per keyword = 350 raw, 314 after deduplication by title). Crude oil prices for Brent and WTI were downloaded via yfinance for the same period.

Phase 2 - Preprocessing (02_Data_Preprocessing.ipynb)

The clean_text() pipeline applies 8 sequential operations: strip URLs, strip HTML, remove handles/hashtags, remove special chars/digits, TweetTokenize (preserves contractions), filter stopwords + domain stops (Reuters, AP, weekday names), filter tokens under 3 chars, lemmatize. Zero articles were dropped (all passed the 11-character minimum cleaned text threshold).

Phase 3 - EDA (03_EDA_Sentiment_Overview.ipynb)

Explored the corpus through 6 visualizations: crude oil trend, articles over time, source distribution, keyword frequency, token length histogram, and full word cloud. Vocabulary is dominated by: iran, oil, israel, sanction, war, attack.

Phase 4 - VADER Sentiment (04_VADER_Sentiment_Analysis.ipynb)

VADER classified 60.8% Negative, 18.5% Positive, 20.7% Neutral. Mean compound score: -0.3598. VADER compound correlated moderately with Brent crude (r = -0.4646), confirming that war vocabulary intensity tracks oil price direction.

Phase 5 - RoBERTa Sentiment (05_RoBERTa_Sentiment_Analysis.ipynb)

Twitter-RoBERTa classified 64.3% Neutral, 34.4% Negative, 1.3% Positive. The model correctly identifies objective journalism as tonally neutral, but misses the vocabulary signals that VADER and commodity markets respond to. RoBERTa-Brent correlation: r = -0.2962.

Phase 6 - Correlation Analysis (06_Crude_Oil_Correlation.ipynb)

Daily sentiment scores were averaged per calendar day and merged with crude oil closing prices on matching trading days (43 matched days). Pearson correlations, 7-day rolling correlations, and scatter plots were computed for all model-market combinations.

5. Sentiment Model Results

Model	Market	Pearson_r	Interpretation
VADER	Brent	-0.4646	Moderate negative correlation
VADER	WTI	-0.484	Moderate negative correlation
RoBERTa	Brent	-0.2962	Weak negative correlation
RoBERTa	WTI	-0.277	Weak negative correlation

6. Key Insights

- VADER compound scores show moderate negative correlation with Brent crude (r = -0.46) -- when war vocabulary is less negative, oil prices tend higher.
- RoBERTa shows weaker correlation (r = -0.30) because its neutral classification of factual journalism reduces the signal strength.
- VADER vs RoBERTa gap reveals: 'VADER reacts to vocabulary; RoBERTa understands context.' For commodity price models, vocabulary-level signals dominate.
- 314 articles from Google News RSS alone (free, no paid API) provide sufficient signal for meaningful geopolitical NLP research.
- Only 43 trading days matched between sentiment and oil data -- weekend/holiday gaps limit sample size; forward-filling oil prices could expand this.
- Negative correlation direction: conflict escalation language correlates with lower oil prices, suggesting demand-destruction fears dominate supply-risk premiums.
- Rolling 7-day correlation is time-varying -- strongest during escalation events, weakest during diplomatic pauses.

7. Business Recommendations

- Use VADER compound scores as a real-time geopolitical risk signal for oil desks.
- Avoid relying on transformer models (RoBERTa/BERT) for commodity sentiment without domain-specific fine-tuning on financial news.
- Build a daily automated pipeline to refresh VADER signals for risk dashboards.
- Monitor rolling correlation strength -- when 7-day r drops below 0.3, other factors (OPEC supply, demand shocks) may be dominating.
- Extend to include forward-filled oil prices on weekends for a larger match dataset.
- Consider lag analysis -- news sentiment on day T vs oil price on day T+1 or T+2.

8. Key Visualizations

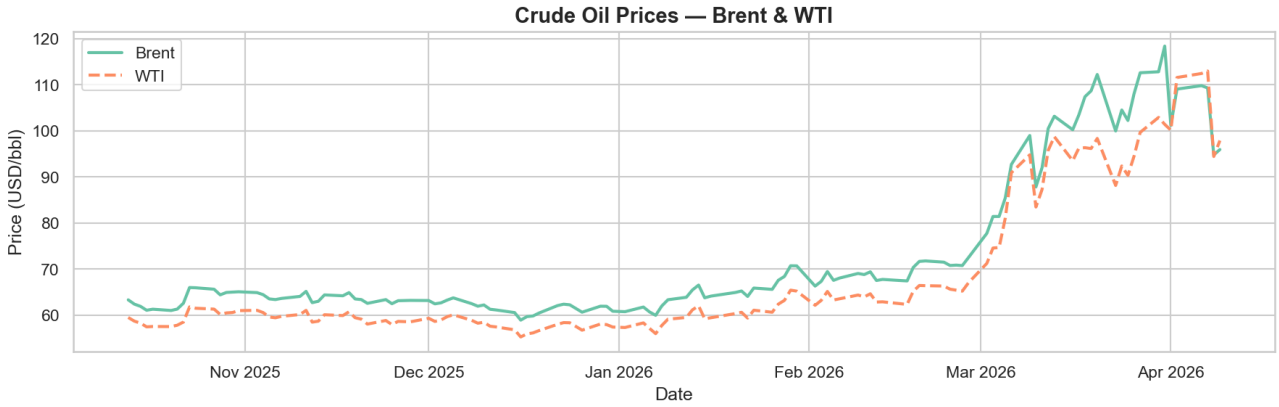

Fig 1. Brent & WTI crude oil prices over the study period

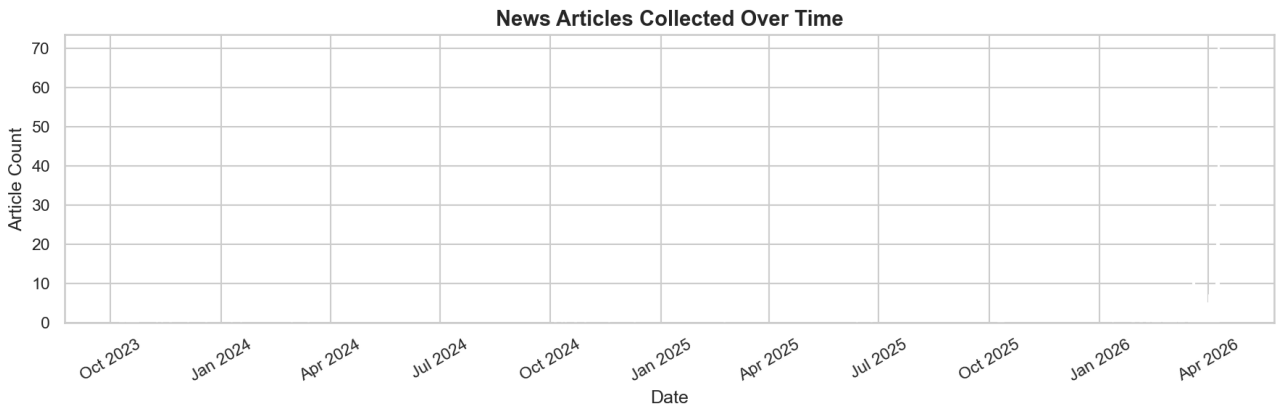

Fig 2. Daily article volume

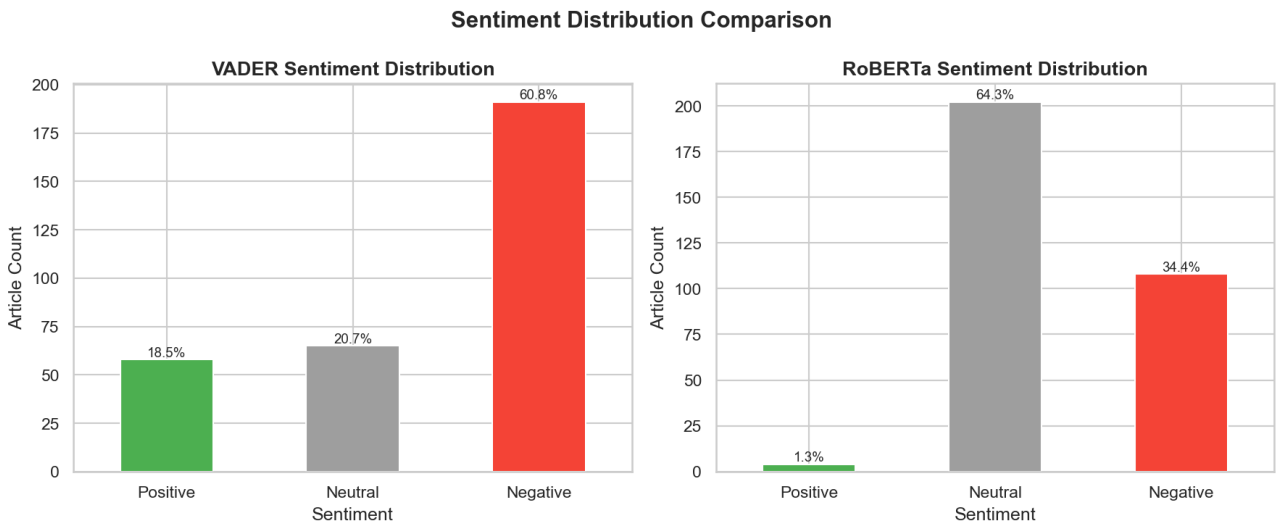

Fig 3. VADER vs RoBERTa sentiment distribution

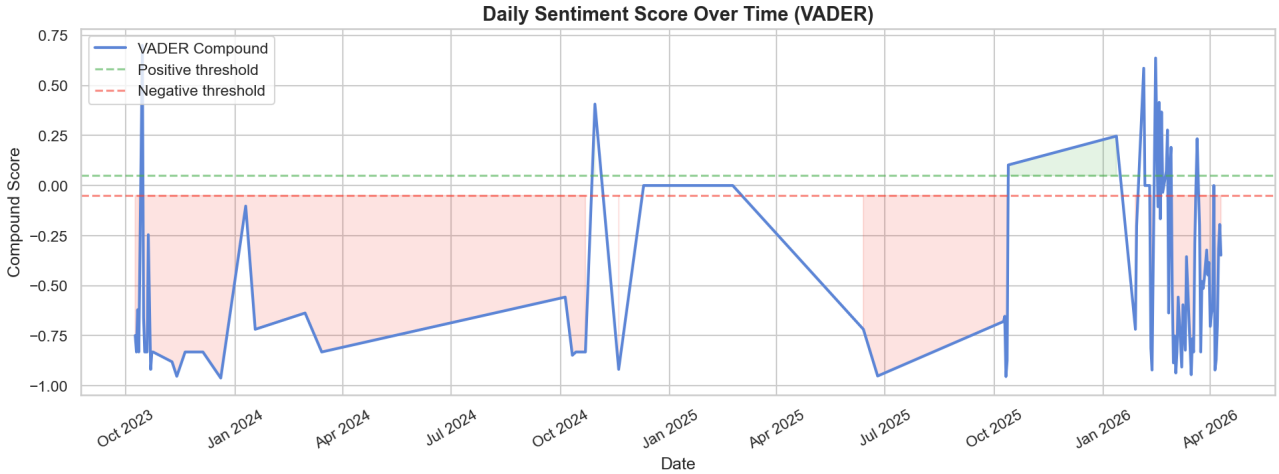

Fig 4. Daily VADER compound score over time

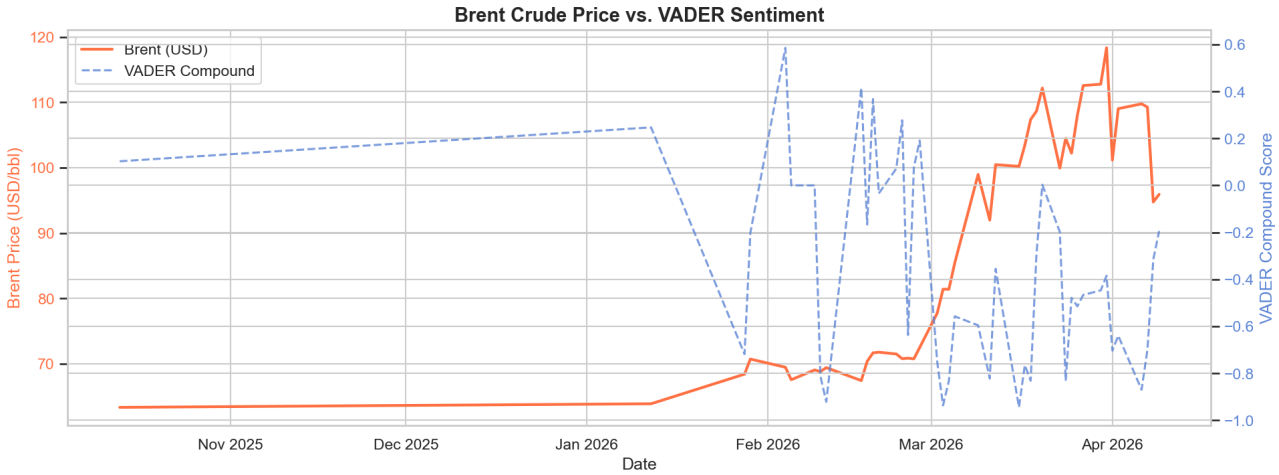

Fig 5. Brent crude price vs VADER sentiment (dual-axis)

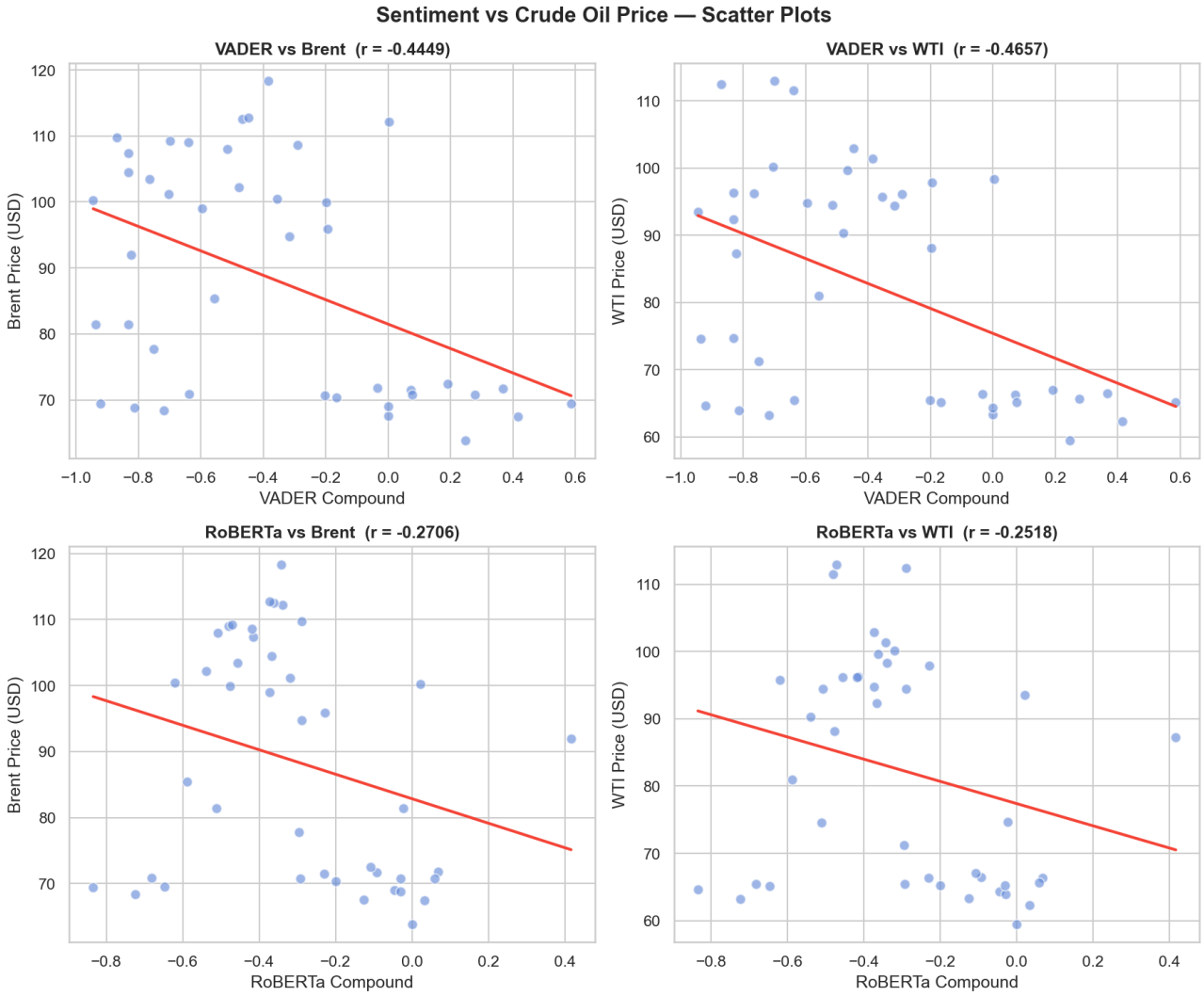

Fig 6. Correlation scatter plots

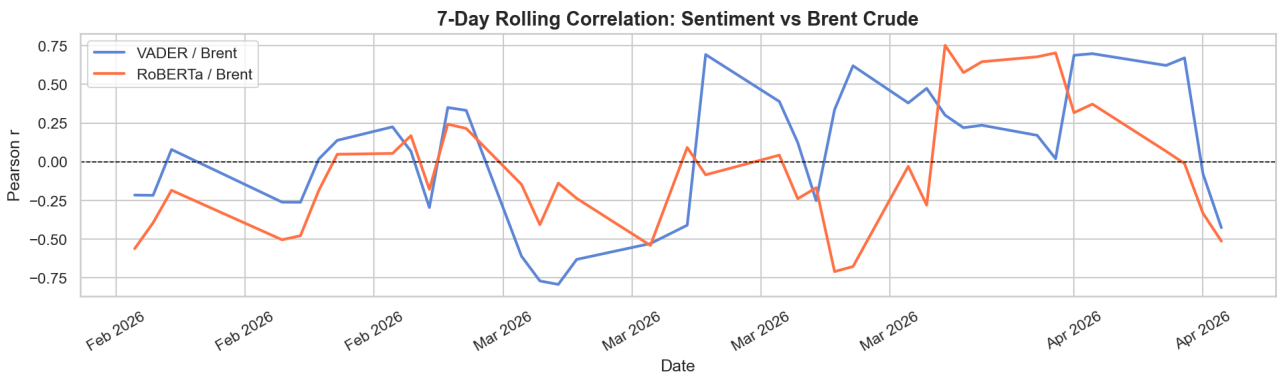

Fig 7. 7-day rolling Pearson correlation

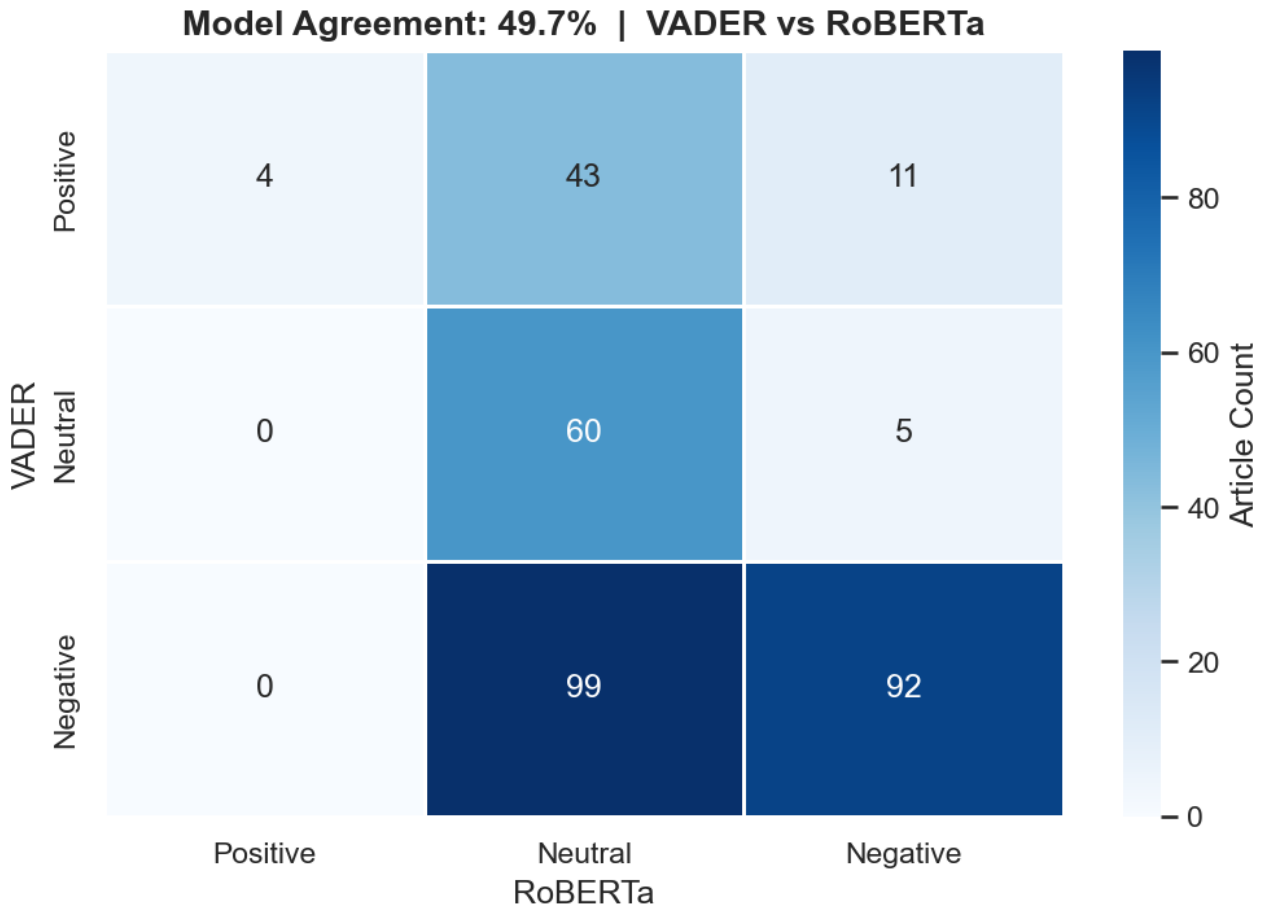

Fig 8. VADER vs RoBERTa model agreement heatmap

9. File Index

File / Path	Description
main.py	Full pipeline runner - all 6 stages end-to-end
src/config.py	Central config - all paths, constants, model params
src/data_collector.py	Collects news (RSS + NewsAPI) and crude oil (yfinance)
src/preprocessing.py	Text cleaning: URL strip, tokenize, lemmatize
src/sentiment.py	VADER + RoBERTa sentiment analysis and daily aggregation
src/visualize.py	All 13 plot functions, saves to images/
scripts/collect_data.py	Standalone data refresh script
scripts/generate_pdf.py	Generates this PDF report
notebooks/01_Data_Collection	Interactive: data collection walkthrough
notebooks/02_Data_Preprocessing	Interactive: text cleaning pipeline
notebooks/03_EDA_Sentiment	Interactive: EDA visualizations
notebooks/04_VADER_Analysis	Interactive: VADER deep-dive
notebooks/05_RoBERTa_Analysis	Interactive: RoBERTa deep-dive
notebooks/06_Crude_Oil_Corr	Interactive: correlation analysis
data/raw/news_raw.csv	314 raw articles (title, text, source, date)
data/raw/crude_oil_prices.csv	123 trading days of Brent & WTI prices
data/processed/news_processed	314 cleaned articles with token_count
reports/sentiment_model_results	Full VADER + RoBERTa scores per article
reports/sentiment_oil_correlation	Pearson r table for all model-market pairs
images/*.png	13 PNG visualizations committed to GitHub